THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A tent is a cylinder of radius 3.5 m and height 4 m surmounted by a cone of the same radius and height 12 m. The area of canvas (in m^2) needed to make the tent is: 1

(a) 88 m^2 **(b)** 225.5 m^2 **(c)** 220 m^2 **(d)** 137.5 m^2

2. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 6, 9, 7, 5, 16, 7 respectively, The mean of the data is: 1

(a) 32.4 **(b)** 35 **(c)** 33.4 **(d)** 270

3. A red die and a blue die are rolled together. The probability that the sum is a multiple of 4 is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{5}{18}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{9}$

4. A hemispherical bowl has outer radius 6 cm and inner radius 3 cm. The volume of material used to make the bowl is: 1

(a) 792 cm^3 **(b)** 396 cm^3 **(c)** 1188 cm^3 **(d)** 198 cm^3

5. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 10, 6, 15, 12, 5, 2 respectively, The median of the data is: 1

(a) 30 **(b)** 20 **(c)** 26 **(d)** 25.4

6. A coin is tossed three times. The probability of getting at least one tail is: 1

(a) $\frac{1}{8}$ **(b)** $\frac{3}{4}$ **(c)** 1 **(d)** $\frac{7}{8}$

7. A hemispherical depression of diameter 21 cm, the largest possible, is scooped out of the top face of a cubical block of edge 21 cm. The volume of the remaining block is: 1

(a) 2425.5 cm^3 **(b)** 9261 cm^3 **(c)** 6835.5 cm^3 **(d)** 4410 cm^3

8. For the data with classes 60–70, 70–80, 80–90, 90–100, 100–110, 110–120, 120–130 with frequencies 2, 16, 10, 2, 4, 5, 11 respectively, The mode of the data is: 1

(a) 94.8 **(b)** 70 **(c)** 77 **(d)** 80

9. 30 cards numbered 1 to 30 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a multiple of 3 or 5 is: 1

(a) $\frac{7}{15}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{8}{15}$ **(d)** $\frac{13}{30}$

10. A solid metallic sphere of radius 12 cm is melted and drawn into a wire of circular cross-section of radius 3 cm. The length of the wire is: 1

(a) 768 cm **(b)** 512 cm **(c)** 256 cm **(d)** 128 cm

11. The mean of a distribution with classes 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 13, 4, p, 8, 14, 5 respectively is 39.2. The value of p is: 1

(a) 5 **(b)** 11 **(c)** 6 **(d)** 8

12. A bag contains 3 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. The value of x is: 1

(a) 2 **(b)** 9 **(c)** 6 **(d)** 5

13. A solid sphere of radius 3 cm is completely dropped into water in a cylindrical vessel of base radius 6 cm, and no water spills. The rise in the water level is: 1

(a) 3 cm **(b)** 0.5 cm **(c)** 1 cm **(d)** 2 cm

14. The median of a distribution with classes 60–70, 70–80, 80–90, 90–100, 100–110, 110–120 and frequencies 18, 18, x, 20, y, 12 respectively, and total frequency 100, is 91.5. The value of x is: 1

(a) 11 **(b)** 12 **(c)** 23 **(d)** 10

15. A bag contains 3 red and 5 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{1}{2}$. The value of x is: 1

(a) 2 **(b)** 1 **(c)** 4 **(d)** 3

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A solid metallic cone of base radius 6 cm and height 3 cm is melted and recast into a solid sphere. Find the radius of the sphere. 2

17. The marks obtained by students in a test are given below. 2

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	6	9	7	5	16	7

Find the mean.

18. A red die and a blue die are rolled together. Find the probability that the sum is a multiple of 3. 2

19. A solid metallic cylinder of base radius 6 cm and height 48 cm is melted and recast into solid spheres of radius 3 cm. Find the number of spheres obtained. 2

20. The marks obtained by students in a test are given below. 2

Marks	20–30	30–40	40–50	50–60	60–70	70–80
Number of students	4	4	10	8	8	6

Find the median.

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A capsule is a cylinder of radius 10.5 cm and length 12 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

22. The marks obtained by students in a test are given below. 3

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	10	6	15	12	5	2

Find the mean.

23. A bag contains 7 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. Find the value of x. 3

24. A toy is a cone of base radius 21 cm and height 14 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

OR

The weights (in kg) of students of a class are given below. 3

Weight (kg)	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	6	9	7	5	16	7

Find the median.

25. A bag contains 3 red and 6 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{1}{2}$. Find the value of x. 3

OR

A wooden article is made by scooping out a hemisphere of radius 3.5 cm from each end of a solid cylinder of height 15 cm and base radius 3.5 cm. Find the total surface area of the article. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A conical vessel of radius 6 cm and height 20 cm is full of water. The water is poured into a cylindrical vessel of base radius 4 cm. Find the height of the water in the cylindrical vessel. 5

OR

The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 and frequencies 6, 9, 7, p, 16, 7 respectively is 32.4. Find the value of p. 5

27. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the product of the numbers is 12,
(ii) the sum is a prime number,
(iii) the sum is a multiple of 3.

OR

Water from a full cylindrical tank of radius 7 cm and height 20 cm is poured into an empty rectangular tank of base 22 cm by 14 cm. Find the height of water in the rectangular tank. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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